

The Cycle-Constrained Aerial Suppression Base and Water Point Location Problem: Joint Siting under Uncertainty with an Application to Cundinamarca, Colombia

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Abstract

Aerial wildfire suppression productivity is governed by a three-leg cycle: an aircraft positions once from its base to the fire, then repeats the fire-to-water refill leg on every drop. The refill leg is paid on every cycle, so water point location, not base-to-fire distance alone, drives how much water a fire actually receives. We introduce the cycle-constrained aerial suppression base and water point location problem, the first model to site aerial bases and water refill points jointly, with aircraft productivity endogenous to the refill cycle, within a two-stage stochastic program whose objective is a mean-risk combination of expected loss and conditional value-at-risk. The bilinear dispatch and refill coupling admits an exact big-M linearization, and a fix-and-optimize matheuristic with large neighborhood search over the first-stage variables makes the full candidate catalog tractable. We build a fully sourced case study for Cundinamarca, Colombia, from open data: 2,413 fire events clustered from satellite detections of calendar year 2024, 120 candidate bases, 5,449 candidate water points, and the Sikorsky S-70i Firehawk that Colombia is acquiring. Three findings stand out. First, under the real program budget the system is fleet-limited: the optimal objective is insensitive to fire growth constants, site costs, and the suppression window across their entire plausible ranges, while the number of affordable aircraft dominates. Second, a best-case sequential planner (bases first, water second) matches the integrated optimum in single-aircraft regimes, and we characterize structurally why; the integrated model's value concentrates where fleets grow or water becomes the binding margin. Third, the direct mixed-integer program hits a tractability cliff precisely in those multi-aircraft regimes, which the matheuristic crosses at negligible cost.

1 Introduction

Colombia's eastern Andean department of Cundinamarca leads the country's historical record of reported forest fires, and the extreme January 2024 fire episode around Bogotá placed aerial suppression at the center of national policy. In response, the Colombian Aerospace Force (FAC), with the national disaster risk management unit (UNGRD), committed 150,000 millones COP to acquire two Sikorsky S-70i Firehawk helicopters, the first of their class in Latin America (Fuerza Aeroespacial Colombiana, 2026). A country buying firefighting aircraft faces an immediate strategic question that current models do not answer: where should the supporting bases and, critically, the water refill points be located, and how should a single budget be split among bases, aircraft, and water infrastructure?

The operational physics of helicopter suppression make this a joint question. An aircraft flies from its base to the fire once, to position, and then cycles between the fire and the nearest usable water point for every drop. As the number of drops grows, total delivered water is dominated by the fire-to-water leg, which is paid on every cycle, not by the base-to-fire leg, which is paid once. Aircraft productivity is therefore endogenous to the joint configuration of

bases and water points. A classical facility location model built on base-to-fire distance alone, of the p -median family, cannot represent this tradeoff, and, as Section 2 shows, no existing model sites bases and water points jointly with cycle-endogenous productivity.

This paper makes four contributions. First, we formulate the cycle-constrained aerial suppression base and water point location problem as a two-stage stochastic program: first-stage decisions open bases, station aircraft, and enable water points under a single budget; second-stage recourse dispatches aircraft to fires scenario by scenario. The objective is a mean-risk convex combination of expected loss and conditional value-at-risk (Rockafellar and Uryasev, 2000). Fire containment is physical: a fire is contained if the liters delivered within a critical window exceed a requirement that grows with arrival time and rate of spread, so no historical suppression outcome data, which do not exist for Colombia, are needed. The only nonlinearity, a bilinear product of dispatch and refill assignment, admits an exact McCormick linearization.

Second, we develop a fix-and-optimize matheuristic (Helber and Sahling, 2010) with large neighborhood search (Shaw, 1998) over the first-stage variables. Each iteration solves a reduced sub-MILP that is exact by construction, over a geographic neighborhood of candidate sites plus the incumbent’s open sites; a probabilistic random destroy operator, standard in adaptive large neighborhood search (Ropke and Pisinger, 2006), provably escapes a class of local optima that purely geographic neighborhoods cannot leave. The reduced sub-problem keeps the full scenario set, so every accepted move is a verified improvement for the true problem.

Third, we construct a fully sourced, replicable case study for Cundinamarca from open data only: NASA FIRMS satellite detections for all of 2024, clustered into events with ST-DBSCAN (Birant and Kut, 2007); land cover, slope, and wind from ESA WorldCover, SRTM, and NASA POWER; population exposure from WorldPop, cross-checked against the Colombian census; 120 candidate bases from the national aerodrome index and 5,449 candidate water points from OpenStreetMap and the regional water authority; and the real Firehawk’s cost, tank, and speed. Every parameter is either traced to a verified source or explicitly declared an assumption and swept in the sensitivity analysis.

Fourth, our computational study yields policy-relevant structural findings rather than only benchmark numbers. Under the real program budget the deployment is fleet-limited: one Firehawk absorbs half the budget, at most one fire per day-scenario can be served, and the optimal objective is flat across the entire plausible ranges of fire growth constants, site costs, and the suppression window, while budget (the number of affordable aircraft) moves it strongly. A best-case sequential planner that sites bases first and water second matches the integrated optimum in every single-aircraft configuration we tested, and we explain structurally why; conversely, the direct MILP hits a tractability cliff exactly where the integrated question becomes interesting, in multi-aircraft regimes, which the matheuristic crosses in seconds.

The remainder of the paper is organized as follows. Section 2 reviews the literature and states the gap. Section 3 formulates the model. Section 4 describes the Cundinamarca case study data. Section 5 presents the matheuristic. Section 6 reports the computational study, and Section 7 concludes with limitations and ongoing work.

2 Related work and gap

Aerial resource location for wildfire suppression began with airtanker base selection: Hodgson and Newstead (1978) chose bases with location-allocation models for one-strike initial attack, and MacLellan and Martell (1996) based airtankers for Ontario with an expected-cost model over historical demand. Modern strategic deployment models keep this base-side focus: Sköld Gustafsson et al. (2024) locate aerial firefighting resources in Sweden with a binary program over wildfire risk data. Stochastic treatments deploy suppression resources across existing stations under fire-occurrence uncertainty: the scenario-based standard-response models of Haight and Fried (2007), the simulation plus stochastic integer programming approach of

Ntaimo et al. (2013), the two-stage seasonal deployment and dispatch design of Wei et al. (2015), which is the closest methodological comparator to our second stage, and the robust prepositioning of Mendes and Alvelos (2025). Recent work also allocates heterogeneous aerial fleets across capacity-constrained airbases under efficiency and equity objectives (Tasias and Nenes, 2026). In all of these, water is either absent or exogenous.

A separate stream sites water infrastructure, with the aircraft side fixed: Ramalho et al. (2021) and Ramalho et al. (2024) allocate water reservoirs by fire risk with geotechnological methods, and Alfaro et al. (2024) optimize reservoir placement for helicopter transport in Valparaíso. Operational models capture the refill cycle but take infrastructure as given: Rodríguez-Veiga et al. (2018) assign aircraft to flight routes and refueling points during suppression, Rodríguez Barreiro et al. (2024) plan helicopter circuits including water loading points for a given fire, and Skorin-Kapov et al. (2024) schedule aerial resources against zone-assigned water loading points. Closest in scope, Maraş et al. (2023) site both heliports and water sources with GIS-based multi-objective programming, but in two separate sequential deterministic models with fixed response-time thresholds, without a refill cycle and without uncertainty.

Table 1 classifies this literature against the three conditions that jointly define the gap: (a) joint siting of bases and water points in one model, (b) aircraft productivity endogenous to the base to fire to water cycle, and (c) strategic, pre-season decisions under uncertainty. Reading the table by column isolates the gap. Cycle-dependent productivity appears only at the operational level, where no siting takes place; joint siting appears in no prior model; and no work combines cycle-dependent productivity with strategic, uncertainty-aware siting. The proposed model is the only row satisfying all three, under stochastic programming with a mean-risk CVaR objective. The nearest works each own a single pillar: Maraş et al. (2023) site both facility types but sequentially, deterministically, and without a cycle; Rodríguez Barreiro et al. (2024) capture the cycle, and even select loading points, but operationally, with given infrastructure; and Wei et al. (2015) are strategic and stochastic but site no water and use coverage rather than a refill cycle.

3 Problem formulation

3.1 Setting and notation

A planner chooses, before the fire season, which candidate aerial bases to open, how many aircraft to station at each, and which candidate water refill points to enable, under a single budget. Fire days then realize randomly. On a fire day, aircraft are dispatched from open bases to fires; an aircraft serving a fire refills at exactly one enabled water point for the whole episode, and the water it delivers within the critical suppression window is determined by the three-leg cycle geometry. A fire whose delivered water reaches a physical requirement is contained; otherwise it escapes and its exposed population counts as loss.

Table 2 summarizes the notation. Distances are in meters in the national planimetric CRS (EPSG:9377), time is in hours, and money is in Colombian pesos (COP).

3.2 Cycle productivity and physical containment

For every i, f, k, m , and s , the following quantities are parameters, computable offline, because every input to them is a parameter under the discrete candidate-site, single-aircraft-type scope:

$$\text{cyc}_{fkm} = 2\tau_{fk}^{\text{fw}} + \delta, \quad (1)$$

$$d_{ifkm} = \max\left(0, \left\lfloor \frac{W - \tau_{if}^{\text{bf}}}{\text{cyc}_{fkm}} \right\rfloor\right), \quad (2)$$

$$L_{ifkm} = d_{ifkm} Q_m, \quad (3)$$

Table 1: Classification of the related literature against the three conditions that define the gap. Bases: the model decides the location of aerial bases or heliports. Water: the model decides the location of water refill points or reservoirs. Joint: both location decisions are made within a single integrated model. Cyclic: aircraft productivity is endogenous to the base to fire to water refill cycle within the decision model. Strategic: infrastructure or deployment decisions taken before the season or the fire day. Uncertainty: treatment of uncertainty (No, deterministic; SP, stochastic programming; RO, robust optimization). Partial indicates partial satisfaction, for example positioning resources at existing bases, or associating water demand by zone without siting refill points.

Reference	Bases	Water	Joint	Cyclic	Strategic	Uncertainty
Hodgson and Newstead (1978)	Yes	No	No	No	Yes	No
MacLellan and Martell (1996) ^a	Yes	No	No	No	Yes	No
Sköld Gustafsson et al. (2024) ^b	Yes	No	No	No	Yes	No
Haight and Fried (2007)	Partial	No	No	No	Yes	SP
Ntairo et al. (2013)	Partial	No	No	No	Yes	SP
Wei et al. (2015)	Partial	No	No	No	Yes	SP
Mendes and Alvelos (2025)	Partial	No	No	No	Yes	RO
Alfaro et al. (2024)	No	Yes	No	No	Yes	No
Ramvalho et al. (2021)	No	Yes	No	No	Yes	No
Ramvalho et al. (2024)	No	Yes	No	No	Yes	No
Rodríguez-Veiga et al. (2018)	No	Partial	No	Yes	No	No
Rodríguez Barreiro et al. (2024) ^c	No	Partial	No	Yes	No	No
Skorin-Kapov et al. (2024) ^d	No	Partial	No	No	No	No
Maras et al. (2023)	Yes	Yes	No	No	Yes	No
This paper	Yes	Yes	Yes	Yes	Yes	SP

^a Minimizes expected cost over multi-season historical demand, without recourse.

^b Conference paper (ISCRAM proceedings).

^c arXiv preprint.

^d The Cyclic cell is confirmed only via a secondary paraphrase of the paywalled primary text (drop counts appear to be exogenous inputs); the remaining cells are confirmed against the openly available abstract.

so L_{ifkm} is the liters one aircraft of type m delivers to fire f when dispatched from base i and refilling at point k : it flies the positioning leg once, then completes as many refill cycles as fit in the remaining window. A base too far to complete even one round trip contributes zero drops. This is the formal core of the cycle insight: τ_{if}^{bf} enters once, through the numerator, while τ_{fk}^{fw} divides the window, so water geometry scales productivity multiplicatively.

Containment is physical. Fire f is contained if delivered liters reach

$$R_f = c A_0 e^{\rho_f t_f^{\text{arr}}}, \quad (4)$$

an exponential area growth form adopted as a phenomenological simplification for tractability. We disclose its provenance precisely: exponential fire area growth is documented in structural fire science (Ramachandran, 1986); wildland-specific growth models are typically elliptical (Van Wagner, 1969) and empirical growth varies by fuel and weather, so Equation (4) is a modeling choice that fits a single scalar ρ_f per fire, not a wildland-validated law. Its constants A_0 and c have no Colombia-specific calibration source and are treated as swept sensitivity parameters throughout (Section 6.5). The arrival time t_f^{arr} is defined exogenously as the travel time from the closest candidate base, $\min_{i \in I} \tau_{if}^{\text{bf}}$, so that R_f remains a pure per-fire parameter independent of the dispatch decision.

Table 2: Notation. Every fire-indexed quantity is scoped to its scenario: a bare f means for $f \in F_s$ within scenario s .

<i>Sets</i>	
I	candidate aerial base sites, index i
K	candidate water refill points, index k
M	aircraft types, index m (a single type in this study)
S	SAA scenario sample, index s , probability p_s
F_s	fires in scenario s , index f
<i>Parameters</i>	
B	total first-stage budget, COP
$c_i^{\text{base}}, c_m^{\text{air}}, c_k^{\text{wat}}$	cost to open base i , acquire one aircraft of type m , enable point k
Q_m	tank capacity of type m , liters per drop
v_m	flight speed of type m , meters per hour
W	critical suppression window length, hours
δ	per-cycle operational overhead (fill plus drop), hours
$\tau_{if}^{\text{bf}}, \tau_{fk}^{\text{fw}}$	one-way base-to-fire and fire-to-water travel times, hours
D_f	value at risk of fire f (population exposure)
ρ_f	rate-of-spread parameter of fire f
t_f^{arr}	detection-to-window arrival time of fire f , hours
A_0, c	initial fire area and liters per square meter to control
α, λ	CVaR confidence level and mean-risk weight
<i>First-stage variables</i>	
$y_i \in \{0, 1\}$	1 if base i is opened
$w_k \in \{0, 1\}$	1 if water point k is enabled
$n_{im} \in \mathbb{Z}_+$	aircraft of type m stationed at base i
<i>Second-stage variables (per scenario s)</i>	
$x_{ifms} \in \mathbb{Z}_+$	aircraft of type m sent from base i to fire f
$z_{fks} \in \{0, 1\}$	1 if fire f refills at water point k
$q_{fs} \geq 0$	liters delivered to fire f
$e_{fs} \in \{0, 1\}$	1 if fire f is not contained
$\ell_s \geq 0$	scenario loss
$\eta \in \mathbb{R}, t_s \geq 0$	CVaR auxiliaries
$u_{ifkms} \geq 0$	linearization auxiliary (Section 3.3)

3.3 Two-stage mean-risk formulation

With $M_m = \lfloor B/c_m^{\text{air}} \rfloor$ a budget-derived fleet bound, the model is:

$$\min (1 - \lambda) \sum_{s \in S} p_s \ell_s + \lambda \left(\eta + \frac{1}{1 - \alpha} \sum_{s \in S} p_s t_s \right) \quad (5)$$

$$\text{s.t. } \sum_{i \in I} c_i^{\text{base}} y_i + \sum_{i \in I} \sum_{m \in M} c_m^{\text{air}} n_{im} + \sum_{k \in K} c_k^{\text{wat}} w_k \leq B, \quad (6)$$

$$n_{im} \leq M_m y_i, \quad \forall i, m, \quad (7)$$

$$z_{fks} \leq w_k, \quad \forall f, k, s, \quad (8)$$

$$\sum_{k \in K} z_{fks} \leq 1, \quad \forall f, s, \quad (9)$$

$$\sum_{f \in F_s} x_{ifms} \leq n_{im}, \quad \forall i, m, s, \quad (10)$$

$$q_{fs} = \sum_{i \in I} \sum_{k \in K} \sum_{m \in M} L_{ifkm} u_{ifkms}, \quad \forall f, s, \quad (11)$$

$$u_{ifkms} \leq x_{ifms}, \quad \forall i, f, k, m, s, \quad (12)$$

$$u_{ifkms} \leq M_m z_{fks}, \quad \forall i, f, k, m, s, \quad (13)$$

$$u_{ifkms} \geq x_{ifms} - M_m(1 - z_{fks}), \quad \forall i, f, k, m, s, \quad (14)$$

$$q_{fs} \geq R_f(1 - e_{fs}), \quad \forall f, s, \quad (15)$$

$$\ell_s = \sum_{f \in F_s} D_f e_{fs}, \quad \forall s, \quad (16)$$

$$t_s \geq \ell_s - \eta, \quad \forall s, \quad (17)$$

$$y_i, w_k, z_{fks}, e_{fs} \in \{0, 1\}, \quad n_{im}, x_{ifms} \in \mathbb{Z}_+, \quad q_{fs}, \ell_s, t_s, u_{ifkms} \geq 0, \quad \eta \text{ free.}$$

The objective (5) is the mean-risk convex combination of expected loss and CVaR in the linear reformulation of Rockafellar and Uryasev (2000); $\lambda = 0$ is risk neutral and $\lambda = 1$ is pure CVaR. Constraint (6) is the single budget covering bases, fleet, and water points, which makes the budget split endogenous. Constraints (7) to (10) link stationing to open bases, refills to enabled points, one refill point per served fire, and dispatch to the stationed fleet. The delivered liters coupling is bilinear in the natural formulation, $q_{fs} = \sum L_{ifkm} x_{ifms} z_{fks}$; constraints (12) to (14) replace each product with the auxiliary u_{ifkms} through a McCormick envelope that is exact, not a relaxation, because z_{fks} is binary and $x_{ifms} \leq M_m$ is valid. Constraint (15) enforces physical containment, and (16) to (17) define scenario loss and the CVaR linking. Scenario probabilities must sum to one; the mean-risk CVaR objective is otherwise unbounded, a subtle failure mode we document because it silently arises when subsampling scenario sets without renormalizing.

Section 6.1 verifies computationally that the linearized model and the literal bilinear program solved with a nonconvex MIQCP solver return identical optimal values on real instances. The auxiliary count $|I| \times |K|$ per fire and scenario is what makes the full 5,449-point water catalog intractable for the direct MILP and motivates the matheuristic of Section 5.

The formulation treats each scenario as a single short episode with no aircraft reuse across fires over time: constraint (10) caps total dispatches per scenario at the stationed fleet. With day-length scenarios and a two-hour window this understates what one aircraft could fly in a full day, a deliberate simplification whose consequences we quantify and discuss in Sections 6.5 and 7.

4 Case study: Cundinamarca, Colombia

All data are open, and every processing decision is scripted in the companion repository. Figure 1 shows the study area.

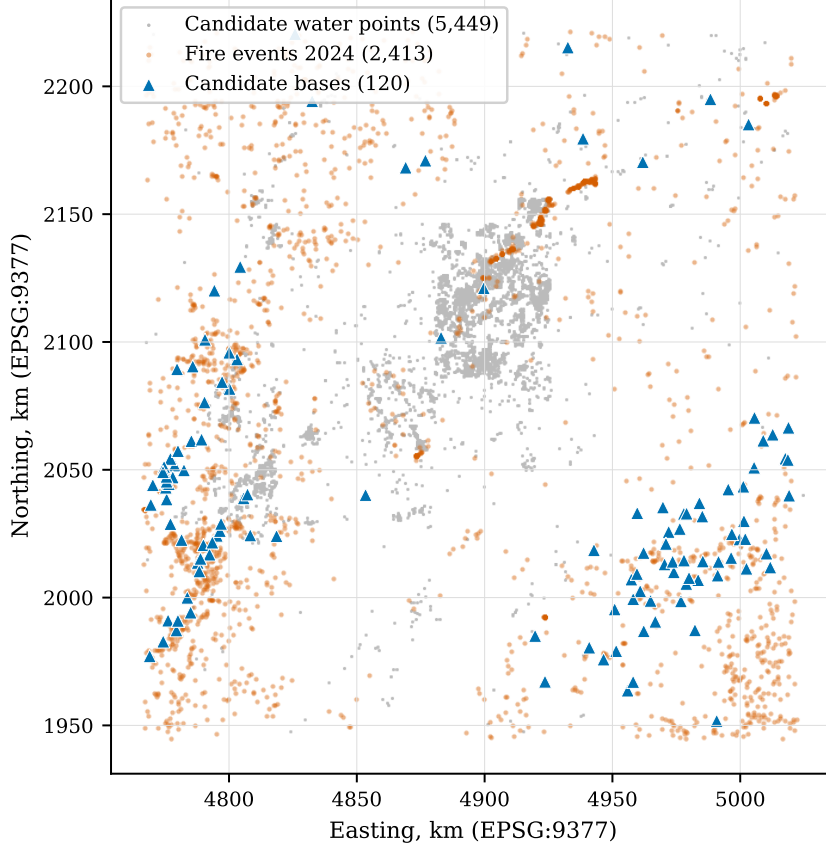


Figure 1: The Cundinamarca instance: 2,413 fire events detected during calendar year 2024, 120 candidate aerial bases, and 5,449 candidate water refill points, in the national planimetric CRS (EPSG:9377).

4.1 Fire events and scenarios

Fire events come from NASA FIRMS VIIRS 375 m active fire detections (NASA FIRMS, 2026) for the full calendar year 2024 over a Cundinamarca bounding box: 4,689 detections pass a nominal-confidence filter and are clustered into 2,413 events with ST-DBSCAN (Birant and Kut, 2007), using $\varepsilon_{\text{spatial}} = 850$ m, anchored on the 375 m sensor resolution, and $\varepsilon_{\text{temporal}} = 1.5$ days, anchored on revisit and persistence; sensitivity of the event catalog to both radii is part of the pipeline. Detections are projected to EPSG:9377, the national single-zone planimetric standard, before clustering.

A scenario is one historical calendar day, matching the stochastic unit of Wei et al. (2015): the 366-day pool of 2024 (282 days with at least one event) is bootstrap-resampled with replacement into $|S| = 200$ equiprobable day scenarios under sample average approximation (Kleywegt et al., 2001). Day scenarios keep the episode structure of constraint (10) valid and assume fleet availability resets daily; cross-day dynamics are not modeled. The satellite record misses cloud-obscured fires and detects active burning rather than ignition, caveats inherited by any FIRMS-based catalog.

4.2 Fire attributes

Each fire’s rate-of-spread parameter follows a land cover base rate modified by slope and wind,

$$\rho_f = \rho_{\text{scale}} \cdot r_{\text{class}(f)} \cdot e^{0.05 \text{slope}_f} \cdot (1 + 0.1 \text{wind}_f), \quad (18)$$

Table 3: Base-case parameters, provenance, and sweep treatment. Citations for each source appear in the accompanying text.

Parameter	Base case	Provenance	Sensitivity
Budget B	150,000 M COP	real Firehawk program spend	swept 80 to 300 G
Window W	2.0 h	NWCG initial attack standard	swept 1, 2, 4, 8 h
CVaR level α	0.95	standard in the CVaR literature	exp. 4 varies risk
Risk weight λ	0.5	midpoint convention	swept 0 to 1
Aircraft cost, tank, speed	Firehawk values	sourced (FAC program)	fixed
A_0	5 ha	no source; sensor-resolution anchor	swept 1 to 30 ha
c	3 L/m ²	no verified source	swept 1.5 to 10
ρ_{scale}	1.0	no source	swept 0.5 to 2
c_i^{base} (uniform)	1,000 M COP	US benchmarks, scaled	swept 300 to 6,000 M
c_k^{wat} (uniform)	50 M COP	US benchmarks, scaled	swept 15 to 300 M

M denotes millones (millions); G denotes thousands of millones (10^9 COP). Agency initial-attack standards genuinely differ, from two-hour containment (National Wildfire Coordinating Group, 2026) to next-morning control (Korkola et al., 2024), hence the window sweep.

with land cover class from ESA WorldCover 10 m (ESA, 2021), slope in degrees from SRTM 30 m (NASA, 2013) by Horn’s method (Horn, 1981), and wind in meters per second from NASA POWER (NASA POWER, 2026) at each fire’s location and date. The relative rates r_{class} are ordinally grounded in the fire behavior fuel models of Anderson (1982), grass fastest, shrub intermediate, timber slower, non-burnable near zero; their absolute magnitude has no Colombian calibration source, so the overall scale ρ_{scale} is a swept sensitivity parameter, and the exponential slope response follows Rothermel (1972) in form with an illustrative coefficient. The value at risk D_f is the WorldPop 2020 constrained population count (WorldPop, 2020) within 1 km of the fire, an unmonetized exposure index cross-checked at municipal level against the 2018 census (DANE, 2018).

4.3 Candidate sites and aircraft

The 120 candidate bases combine the national civil aerodrome index (Aerocivil, 2025) with military installations; the 5,449 candidate water points combine named OpenStreetMap water bodies (OpenStreetMap contributors, 2026) with the regional water authority’s lagoon layer (CAR Cundinamarca, 2026). The aircraft is the Sikorsky S-70i Firehawk that Colombia is acquiring (Fuerza Aeroespacial Colombiana, 2026): 75,000 millones COP per unit (the program price of 150,000 millones for two), a 1,000 gallon (3,785 liter) tank, snorkel refill in under one minute from water at least 45 cm deep, which is what makes lagoon-scale water points usable, and a 150 knot cruise (277.8 km per hour). The per-cycle overhead δ is the sourced sub-minute refill plus an assumed 30 second drop and positioning allowance.

4.4 Parameter decisions and disclosed assumptions

Table 3 lists the base-case values of the governing parameters and the provenance of each. Three are evidence-anchored single values; site costs and the fire growth constants have no Colombia-specific source and are declared illustrative sweep ranges, never calibrated values, a disclosure discipline we consider a methodological contribution of the case study.

5 Solution method

The direct MILP is tractable only for truncated candidate sets: the linearization adds $|I| \times |K|$ auxiliaries per fire and scenario, over 650,000 per fire-scenario pair at the full catalog. We therefore solve the full problem with a fix-and-optimize matheuristic (Helber and Sahling,

2010), a MIP-based improvement scheme distinct from local branching (Fischetti and Lodi, 2003), with large neighborhood search (Shaw, 1998) over the first-stage variables.

The incumbent is a first-stage solution (y, w, n) . Each iteration selects a neighborhood of candidate sites to free, builds a reduced model containing only the freed sites plus the incumbent’s open sites, fixes the non-freed open sites at their incumbent values, solves the reduced sub-MILP exactly over the full scenario set, and accepts the result if it strictly improves the incumbent. Dropping a site that is closed in the incumbent and outside the neighborhood is exact, not an approximation: such a site is mathematically identical to one with its binary fixed at zero. Every accepted iteration is therefore a verified improvement of the true objective, and the sub-MILP size is bounded by the neighborhood, not the catalog.

Neighborhood selection follows Shaw’s relatedness principle in geographic form: a random seed site anchors each iteration, and the nearest candidate bases and water points to it are freed together, matching the problem’s own coupling, since sites an aircraft would cycle between are exactly the sites worth reconsidering jointly. Pure geographic neighborhoods, however, provably admit local optima that no number of iterations escapes: when the only improving move must close an open base and open a distant one within the same iteration, for example when the budget affords a single aircraft so relocation is the only option, no single seed’s nearest-sites neighborhood ever contains both. We observed exactly this on a real instance, where the optimal base lay 174 km from the incumbent’s and 200 geographic iterations produced no improvement. Following standard adaptive large neighborhood search practice (Ropke and Pisinger, 2006), each iteration is therefore, with probability π , a random destroy that frees sites drawn uniformly from the whole catalog; unit tests reproduce the trap instance and confirm the random operator escapes it reliably across seeds. A parameter sweep on a real instance with a known optimum (24 configurations, five seeds each) found quality saturated at the base case, all configurations reached the optimum, so neighborhood sizes are chosen for runtime (five bases, ten water points per iteration) and $\pi = 0.3$ is retained as insurance against the documented trap regime.

6 Computational study

All experiments run on an Intel Core i7-13620H with 16 GB RAM under Windows 11, with Gurobi 13.0.2 as the MIP solver (PuLP 3.3.2 as the modeling layer). Unless stated otherwise, instances use the base-case parameters of Table 3, candidate subsets are taken in catalog order, and scenario subsets renormalize probabilities. Every table and figure is generated by a committed script from a committed results file.

6.1 Exactness of the linearization

Table 4 compares the linearized MILP against the literal bilinear formulation, solved by Gurobi as a nonconvex MIQCP, on real instances of increasing size. Optimal objectives match exactly in every case, confirming computationally that the McCormick envelope is exact. At these sizes the bilinear arm is somewhat faster, so we make no speed claim for the linearization; the presolve of a modern solver evidently linearizes the binary product internally at least as well. The linearized model remains the formulation of record because it is solver-portable and is the sub-problem the matheuristic solves.

6.2 Tractability and the matheuristic

Figure 2 shows the tractability cliff. With one affordable aircraft the direct MILP at 20 bases, 50 water points, and 10 scenarios (81 fires) solves to proven optimality in under five minutes. Raising the budget into the two-aircraft regime on the same instance, the direct solve ran

Table 4: Experiment 1: linearized MILP versus literal bilinear formulation, two real scenarios (15 fires), base-case parameters. Objectives are identical to solver precision at every size.

$ I $	$ K $	MILP obj.	MILP s	Bilinear obj.	Bilinear s	Match
3	5	73.847	0.03	73.847	0.02	yes
5	10	73.847	0.09	73.847	0.05	yes
8	15	73.847	0.22	73.847	0.16	yes

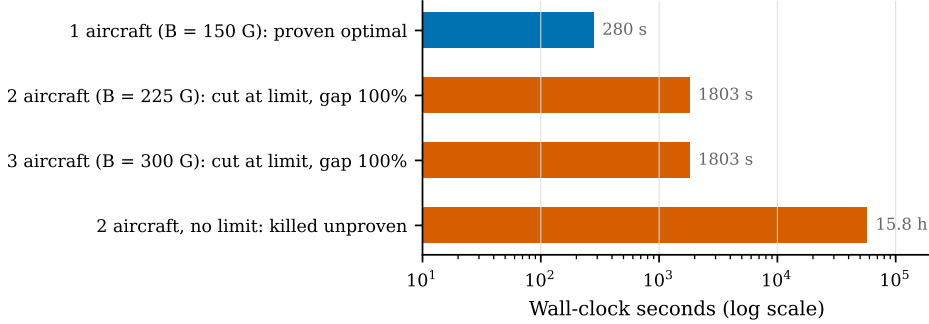


Figure 2: Direct integrated MILP wall-clock time at 20 bases, 50 water points, 81 fires, as the fleet regime grows. Blue: proven optimal. Vermillion: not proven (cut at the 1,800 s limit with 100 percent MIP gap, or terminated unproven).

for almost 16 wall-clock hours (about 48 CPU-hours) without proving optimality before being terminated; re-run under a 1,800 second limit it returned an incumbent with a 100 percent MIP gap, that is, a lower bound near zero. The combinatorics of jointly placing two aircraft, their bases, and shared water investments is qualitatively harder, and this is precisely the regime where the integrated question matters.

The matheuristic crosses the catalog-scale barrier that the direct model cannot approach at all. On the full 120-base, 5,449-water-point catalog with two real scenarios it converged in 2.5 seconds (Gurobi sub-solves; 36 seconds with the open-source CBC solver) to the same objective value, 73.847, that the direct MILP proves optimal on truncated instances of the same scenarios, a genuine cross-check of the reduced-sub-MILP exactness argument at real scale. A ten-scenario (81 fire) full-catalog run took 15.8 seconds per iteration. Figure 3 reports the runtime scaling of the sub-solves with neighborhood size from the tuning sweep; water neighborhood size dominates cost, consistent with the $|I| \times |K|$ auxiliary count.

6.3 Integrated versus sequential siting

The star comparison asks what joint optimization is worth against a sequential planner who sites bases first and water second. Our sequential baseline is deliberately strong. Phase A solves the same model with a water-blind relaxation, every fire is assumed to reach its best-case water point in the full catalog at zero cost, under a budget share ϕB for bases and fleet; phase B fixes phase A’s bases and fleet and solves the full model, real water catalog and costs, real total budget, deciding only water points and recourse. We sweep the split ϕ and report the best sequential result, so the comparison is against a best-case sequential planner, not a straw man. By construction the sequential solution is feasible for the integrated model, so it can never win; on a hand-built instance in the companion test suite it is strictly worse at every ϕ , because a water-blind phase A commits its single affordable aircraft toward a marginally more valuable fire whose only water point is economically out of reach, a misdirection no budget split can undo.

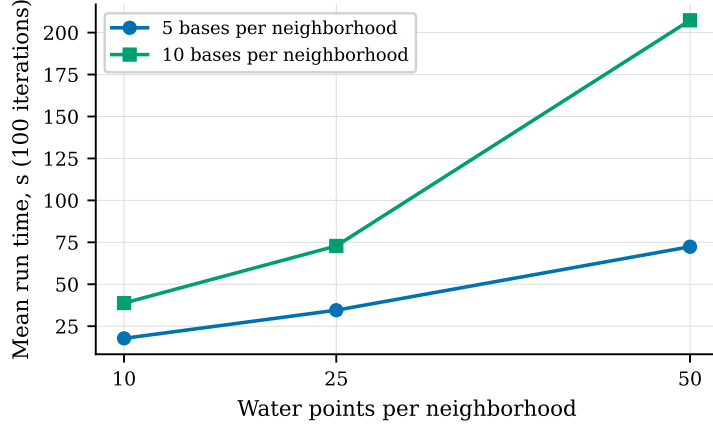


Figure 3: Matheuristic runtime versus neighborhood size (mean of 100 iteration runs across destroy probabilities and five seeds each, 20 base, 50 water point instance).

On the real Cundinamarca instances the result is an honest null with a sharp structural explanation. Across every single-aircraft configuration tested (budgets of 78, 80, 90, and 150 G COP, water costs at both 50 M and the top-of-range 300 M, two and ten scenarios, up to 20 bases and 50 water points), the best sequential solution matches the integrated optimum to solver precision. The reason is regime structure: with $c^{\text{air}} = 75$ G, any budget in $[76, 152)$ G fields exactly one aircraft; constraint (10) then serves at most one fire per scenario, so at most $|S|$ distinct water points are ever needed, and even the tightest leftover budget buys them. Water cannot compete for budget, and a water-blind base choice is as good as an informed one. In the two-aircraft regime (160 G) the integrated arm cannot prove optimality (Section 6.2), so the gap there remains bounded rather than measured: notably, the sequential baseline found, in 26 seconds, a solution matching the incumbent that 30 minutes of direct branch and bound produced, which recommends it independently as a warm start. The honest headline is conditional: sequential siting is near-optimal in single-aircraft regimes, and the integrated model’s value concentrates where fleets grow or water becomes the binding margin, a characterization we consider more useful to a planner than a gap manufactured at a contrived parameter point.

6.4 Risk attitude

Figure 4 sweeps the mean-risk weight λ on the 20-base, 50-water, 10-scenario instance. The efficient frontier is degenerate: expected loss and CVaR are both flat from $\lambda = 0$ to 0.75, because a single affordable aircraft cannot hedge the worst day at all, there is no tail-protection to buy. At $\lambda = 1$, pure CVaR, the solver returns a dominated plan: identical CVaR, 35 percent worse expected loss, three more escaped fires, since a pure tail objective is indifferent to everything off the tail. This is the textbook argument for the mean-risk combination over pure CVaR, reproduced on real data. With ten equiprobable scenarios, $\text{CVaR}_{0.95}$ equals the worst-scenario loss, the expected small-sample behavior under SAA, which a full replication analysis (ongoing work) will quantify with optimality-gap confidence intervals.

6.5 Sensitivity: a fleet-limited system

A one-at-a-time sweep around the base case varies seven axes: budget, window, A_0 , c , ρ_{scale} , and both site costs, 21 exact solves in total (multi-aircraft budget points are bounded at a 1,800 s limit and reported as incumbents). The result is stark (Figure 5): budget is the only axis that moves the objective. Window from 1 to 8 hours, initial area from 1 to 30 hectares, water

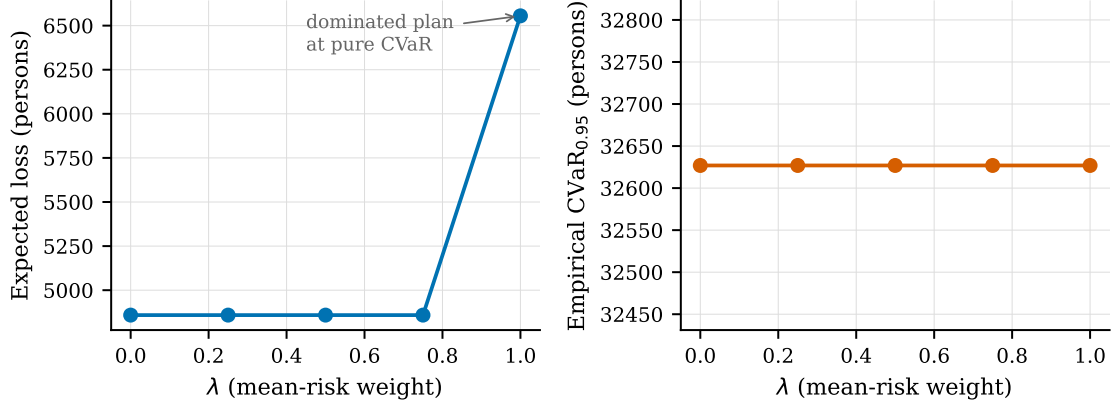


Figure 4: Experiment 4: risk-attitude sweep. Left: expected loss. Right: empirical CVaR_{0.95}, recomputed identically for every λ from each solution’s loss distribution. The frontier is degenerate under a single affordable aircraft, and pure CVaR ($\lambda = 1$) yields a dominated plan.

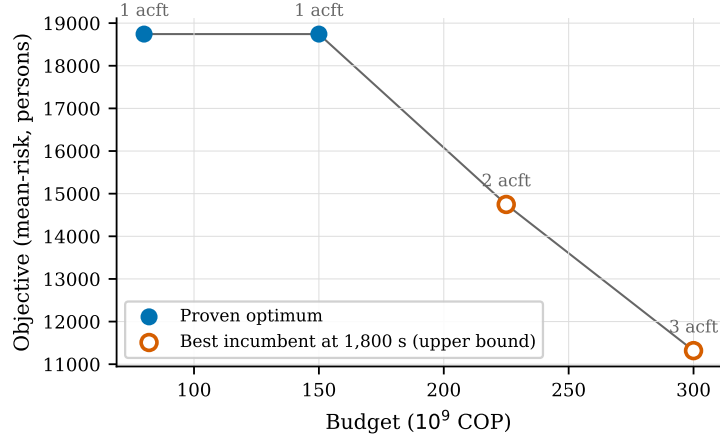


Figure 5: Experiment 6, budget axis: the only sensitivity axis that moves the optimal objective. Annotations give the fleet size fielded. Open markers are 1,800 s incumbents (upper bounds on the optimum).

demand from 1.5 to 10 liters per square meter, spread rates halved or doubled, base costs from 300 to 6,000 M and water costs from 15 to 300 M all leave the optimal objective identical to floating-point precision; only the specific alternate-optimal site choices shuffle. Under the real program budget the system is fleet-limited, not fire-physics-limited and not site-cost-limited: 74 of 81 fires escape because one aircraft serves one fire per day-scenario, so the constants of the containment model have no room to matter. This finding is convenient for the model’s credibility, the policy conclusion is insensitive to exactly the constants we could not calibrate, and it carries a policy edge: at these prices, marginal pesos buy more protection as aircraft (with their basing) than as any refinement of the ground network, until the fleet grows enough for water siting to bind. It must be read alongside the disclosed no-intra-day-reuse simplification, which understates single-aircraft capacity and correspondingly overstates the marginal value of the second aircraft.

7 Conclusions, limitations, and ongoing work

We introduced the first model that jointly sites aerial suppression bases and water refill points with cycle-endogenous aircraft productivity, under a two-stage stochastic program with a mean-risk CVaR objective, gave it an exact linearization and a verified-exact fix-and-optimize matheuristic, and built a fully sourced open-data case study for Cundinamarca around Colombia’s real Firehawk acquisition. The computational study’s contribution is structural: a fleet-limited regime in which sequential siting is provably adequate and the containment constants are immaterial, a sharp tractability cliff exactly where fleets grow, and CVaR degeneracies that argue for the mean-risk form on real data.

Limitations are disclosed throughout and summarized here. The containment law is a phenomenological exponential with uncalibrated, swept constants; scenarios cover a single year’s bootstrap pool, so interannual variability is unrepresented; each scenario is a no-reuse episode, understating per-aircraft daily capacity; the aircraft type is single; and site costs are uniform sweep points. Two pieces of the full experimental program are ongoing: SAA replication with optimality-gap confidence intervals, and measuring, rather than bounding, the integrated-versus-sequential gap in multi-aircraft regimes with the matheuristic as the integrated arm.

Reproducibility. All code, the data pipeline, every experiment script, and the verification log of every decision and citation are available in the companion repository; raw data are redistributable open sources referenced above.

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